

Essence of Galois Theory

A compact and minimalistic notebook on Galois Theory.

Made to collect in a single place all the very essential concepts and results that are needed to prove the non existence of the quintic-formula by radicals.

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1 Algebra remarks

Abstract algebra is a branch of mathematics that studies algebraic structures such as groups, rings, fields, and modules. These structures generalize the familiar algebraic concepts of numbers and operations, allowing mathematicians to explore deeper properties and relationships. Abstract algebra has applications in various areas of mathematics, computer science, physics, and engineering. In this section, we will introduce the fundamental concepts and structures of abstract algebra, along with their significance and applications.

1.1 Operations

"Operation" is just a fancy term to refer functions in the context of algebraic structures. An operation can be either internal or external. An internal operation is a function that takes some elements from the set and returns an element from the set. Internal operations are the most common type of operations in algebraic structures.

Definition 1. *An internal binary operation $\perp$ over a set S is a function $\perp: S \times S \rightarrow S$*

Definition 2. *An external binary operation $\cdot$ over the pair of sets K, V is a function defined as $\cdot: K \times V \rightarrow V$, note that this definition encompasses the case where $K = V$ hence its a generalization of the internal binary operation.*

Definition 3. *An unary operation $!$ over a set S is a function $!: S \rightarrow S$, note that an unary operation can never be external so specifying that it is internal is redundant.*

Definition 4. *With respect to a binary internal operation $\perp$ over a set S , an element $\epsilon \in S$ is said to be the left-identity-element of the operation $\perp$ if $\forall x \in S: \epsilon \perp x = x$.*

Definition 5. *With respect to a binary internal operation $\perp$ over a set S , an element $\delta \in S$ is said to be the right-identity-element of the operation $\perp$ if $\forall x \in S: x \perp \delta = x$.*

Given these very few definitions we can already prove some interesting statements about the properties of the identity elements of an operation. In particular, we can prove that if a set has both a left-identity-element and a right-identity-element, then they are the are both simultaneously the left-identity-element and the right-identity-element of the operation. We can also prove that under these circumstances the two elements are not distinct, meaning that indeed there can only be one identity element for a given operation both on the left and on the right. This property is very important in the study of algebraic structures, as it allows us to define the concept of identity element in a more general way.

Lemma 1. *If two elements $\epsilon, \delta \in S$ exist such that they are left-identity-element and right-identity-element of the operation respectively, then $\epsilon = \delta$.*

Proof. Let $\epsilon \in S$ be the left-identity-element $\perp$, then let $\delta \in S$ be the right-identity-element of $\perp$. The following statements are then true by the very definition of left and right identity elements:

$$\begin{aligned}\forall x \in S : \epsilon \perp x &= x \\ \forall x \in S : x \perp \delta &= x\end{aligned}$$

By the definition of the right-identity-element, we can substitute $x = \epsilon$. By the definition of the left-identity-element, we can also substitute $x = \delta$. Ending up with:

$$\begin{aligned}\epsilon \perp \delta &= \epsilon \\ \epsilon \perp \delta &= \delta\end{aligned}$$

By the transitive property of equality, we can then conclude that $\epsilon = \delta$. Note that this means that the left-identity-element and the right-identity-element are the same element, and that the element itself is therefore the identity element of the operation $\perp$ on both sides (this holds true regardless of whether $\perp$ has any other property or not).

$$\epsilon = \epsilon \perp \delta = \delta$$

□

Definition 6. *With respect to a binary internal operation $\perp$ over a set S , an element $\hat{a}$ is said to be the left-inverse-element of an element $a \in S$ if $\hat{a} \perp a = \epsilon$ where ϵ is the left-identity-element of the operation $\perp$.*

Definition 7. *With respect to a binary internal operation $\perp$ over a set S , an element $\bar{a}$ is said to be the right-inverse-element of an element $a \in S$ if $a \perp \bar{a} = \delta$ where δ is the right-identity-element of the operation $\perp$.*

With these definitions we can now go ahead and prove an interesting property of the inverse elements of an operation. In particular, we can prove that if an element has both a left-inverse-element and a right-inverse-element, then the two inverse-elements are the same under the assumption that the operation is associative, we can also prove that the inverse element is unique.

Please note how this also automatically implies that the identity element is unique since the very existence of both a left and a right inverse element implies that there exists both a left and a right identity element, which itself implies that the two elements are the same and therefore unique.

Lemma 2. *if an element has both a left-inverse-element and a right-inverse-element with respect to an associative operation, then the two inverse-elements are the same.*

Proof. Let $\hat{x}$ be the right-inverse-element of x and $\bar{x}$ be the left-inverse-element of x with respect to an associative operation $\perp$.

$$\begin{aligned}\hat{x} &= \epsilon \perp \hat{x} \\ &= (\bar{x} \perp x) \perp \hat{x} \\ &= \bar{x} \perp (x \perp \hat{x}) \\ &= \bar{x} \perp \epsilon \\ &= \bar{x}\end{aligned}$$

□

Definition 8. *An element $\theta \in S$ is said to be the left-absorbing-element of a binary internal operation $\perp$ if $\forall x \in S : \theta \perp x = \theta$.*

Definition 9. *An element $\zeta \in S$ is said to be the right-absorbing-element of a binary internal operation $\perp$ if $\forall x \in S : x \perp \zeta = \zeta$.*

Lemma 3. *If a given binary internal operation $\perp$ has both a left-absorbing-element and a right-absorbing-element, then the two elements are the same.*

Proof. Let θ be the left-absorbing-element and ζ be the right-absorbing-element of the operation $\perp$ for a given binary internal operation $\perp$. Then the following statements are true by the very definition of left and right absorbing elements:

$$\begin{aligned}\forall x \in S : \theta \perp x &= \theta \\ \forall x \in S : x \perp \zeta &= \zeta\end{aligned}$$

By the definition of the right-absorbing-element, we can substitute $x = \theta$. By the definition of the left-absorbing-element, we can also substitute $x = \zeta$. Ending up with:

$$\begin{aligned}\theta \perp \zeta &= \theta \\ \theta \perp \zeta &= \zeta\end{aligned}$$

By the transitive property of equality, we can then conclude that $\theta = \zeta$. Note that this means that the left-absorbing-element and the right-absorbing-element are the same element, and that the element itself is therefore the absorbing element of the operation $\perp$ on both sides (this holds true regardless of whether $\perp$ has any other property or not).

$$\theta = \theta \perp \zeta = \zeta$$

□

1.2 Algebraic Structures

In this chapter we will define the algebraic structures that we will be using in the rest of the note. In particular, we will define Monoids, Groups, Rings and Fields.

Definition 10 (Monoid). *A monoid $(M, \perp)$ is a set M equipped with an internal binary operation $\perp$ that is both associative, and has some identity element ϵ .*

Definition 11 (Group, Abelian Group). *A group $(G, \perp)$ is monoid for which every element has an inverse. A group is abelian if the operation $\perp$ is commutative.*

Lemma 4. *If $(G, \perp)$ is a group, then, for every element $a \in A$, the identity $k \perp k = k$ implies that a is the identity element.*

Proof.

$$\begin{aligned} a \perp \epsilon &= a \\ a \perp (k \perp k^{-1}) &= a \\ a \perp (k \perp k \perp k^{-1}) &= a \\ a \perp k \perp (k \perp k^{-1}) &= a \\ a \perp k \perp \epsilon &= a \\ a \perp k &= a \end{aligned}$$

□

Lemma 5 (Cancellation-law). *If $(G, \perp)$ is a group, then, $f(x) = x \perp k$ is injective.*

Proof.

$$\begin{aligned} x_1 \perp k &= x_2 \perp k \\ x_1 \perp k \perp k^{-1} &= x_2 \perp k \perp k^{-1} \\ x_1 &= x_2 \end{aligned}$$

□

Definition 12 (Ring). *A ring $(A, \perp, \nabla)$ is a set A equipped with two internal binary operations $\perp$ and ∇ such that $(A, \perp)$ is an abelian group, (A, ∇) is a monoid, and the two operations are compatible in the sense that for all $a, b, c \in A$:*

$$\begin{aligned} a \nabla (b \perp c) &= (a \nabla b) \perp (a \nabla c) \\ (a \perp b) \nabla c &= (a \nabla c) \perp (b \nabla c) \end{aligned}$$

Lemma 6 (Absorbing element). *If $(A, \perp, \nabla)$ is a ring, then the identity element of $(A, \perp)$ is the absorbing element of (A, ∇) . Meaning that an absorbing element always exists.*

Proof. Assuming θ to be the identity element of $(A, \perp)$, we have that the following holds:

$$a \nabla \theta = a \nabla (\theta \perp \theta) = (a \nabla \theta) \perp (a \nabla \theta)$$

□

Definition 13 (Field). *A field $(F, \perp, \nabla)$ is a field for which every element has an inverse with respect to ∇ except the absorbing element, meaning that $(F - \{\theta\}, \nabla)$ is a group.*

Example 1. An example of commutative monoid is $(\mathbb{N}, +)$, where the $+$ operation is the classical sum of natural numbers. The identity element is 0, and the operation is associative. There are no inverse elements for non-zero numbers such that their sum is 0.

Example 2. Another example of commutative monoid is $(P(S), \cup)$, where $P(S)$ is the power set of a set S , and the $\cup$ operation is the union of sets. The identity element is the empty set, and the operation is associative. There are no inverses.

Example 3. An example of non-commutative monoid is $(\{ f : S \rightarrow S \}, \circ)$, where the $\circ$ operation is the function composition. The identity element is the identity function. Not all functions are invertible.

Example 4. Another example of commutative monoid is $(P(S), \cap)$, where $P(S)$ is the power set of a set S , and the $\cap$ operation is the intersection of sets. The identity element is the set S , and the operation is associative. There are no inverses.

Example 5. An example of non-abelian group is $(\mathbb{R}_{2,2}, \times)$, where $\mathbb{R}_{2,2}$ is the set of two by two matrices with real coefficients, and the $\times$ operation is the matrix product. The operation is associative, there is an identity element, and the inverse of a matrix always exists. However, the operation is not commutative.

Example 6. An example of non-abelian group is the group $(B(S), \circ)$ of all bijective functions from a set to itself under the operation of function composition. The inverse is the identity function and the operation is associative. Bijections are invertible.

Example 7. An example of abelian group is $(\mathbb{Q}, \cdot)$, where the $\cdot$ operation is the classical multiplication of rational numbers. The identity element is 1, and the operation is associative. The inverse of x is $\frac{1}{x}$.

Example 8. An example of ring is $(\mathbb{Z}, +, \cdot)$, where the $+$ operation is the classical sum of integers, and the $\cdot$ operation is the classical multiplication of integers.

Example 9. Another ring is $(\mathbb{R}[x], +, \cdot)$ where $\mathbb{R}[x]$ is the set of polynomials with real coefficients, and the $+$ and $\cdot$ operations are the classical sum and of polynomials. The reason why this is not a field is that the polynomial x does not have a multiplicative inverse such that $x \cdot f(x) = 1$. In general, $F[x]$ where F is a field, forms a ring.

Example 10. Canonical examples of fields are $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$.

2 Fundamentals of group theory

In this section, we will explore the concepts of advanced group theory that are related to Galois theory. In particular, we will focus on the symmetric group, the alternating group, and the concept of normal-subgroup and quotient-groups. We will define what does it mean for a group to be Galos-solvable and we will prove that the so called symmetric group S_n is not solvable for $n \geq 5$. This will later be linked to the fact that there are polynomial equations of degree $n \geq 5$ that are not solvable in radicals.

2.1 Permutations, symmetries and cycles

A permutation, is a bijective function from a set to itself. The set of all permutations of a finite set A is a group under the operation of composition. We will see how this group is an extremely important object in Galois theory, as it is used to study the symmetries of the roots of polynomial equations.

Definition 14 (Permutation). *A permutation of a set A is a bijective function $\sigma : A \rightarrow A$.*

Definition 15 (Symmetric group). *The set S_n is the set of all the permutations of the set $\{1, 2, \dots, n\}$. Please notice that the such set is indeed isomorphic to the set of all permutations of the set A , for every choice of A such that it has finite cardinality n .*

We can indicate a permutation σ in S_n with the following notation (matrix notation):

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \dots & \sigma(n) \end{pmatrix}$$

Definition 16 (Cycle). *A cycle of length k is a permutation σ such that every element gets mapped onto itself after k mappings (Repeatedly applying the permutation σ k times, can be written as σ^k since in a group the operation is typically seen as multiplication-like):*

$$\sigma^k(n) = n$$

We can indicate a cycle τ of length k with the following notation (tuple notation):

$$\tau = (a_0, a_1, \dots, a_k)$$

Which represents the permutation described by the following case-by-case definition:

$$\tau(x) = \begin{cases} a_{i+1} & \text{if } x = a_i \wedge x \neq a_k \\ a_1 & \text{if } i = k \\ x & \text{otherwise} \end{cases}$$

Notice that (a_0) can be written as (a_0, a_0) , and that's the identity in every S_n .

Lemma 7. S_n forms the group $(S_n, \circ)$ under the operation of composition.

Proof. It's clearly closed, since the product of two permutations is a permutation. The identity permutation is the one that maps every element to itself, and since every permutation is a bijection, it has an inverse. Also, by the definition of composition, the operation is associative. Thus, S_n is a group under the operation of composition. $\square$

Lemma 8. Given $S_n \subseteq S_{n+1}$, we can express every permutation in S_{n+1} as the product of a permutation in S_n and a length-2-cycle in the form $(n+1, i)$, $1 \leq i \leq n$.

Proof. Let's denote the length-2-cycle as $\beta_i = (n+1, i)$. There exist then exactly n different functions $f_i(\sigma) = \sigma \circ \beta_i$, where i is less than or equal to n .

Now, observe that the function f is injective, since the cancellation law holds for the group of permutations just like any other group (lemma ??).

$$\begin{aligned} f_i(\sigma) = f_i(\tau) &\Rightarrow \sigma \circ \beta_i = \tau \circ \beta_i \\ &\Rightarrow \sigma \circ \beta_i \circ \beta_i^{-1} = \tau \circ \beta_i \circ \beta_i^{-1} \\ &\Rightarrow \sigma = \tau \end{aligned}$$

Also notice that the images of the function f_i are all distinct since they all map to element that have been multiplied by a different factor which is not in the domain. Thus:

$$\left| \bigcup_{1 \leq i \leq n} \vec{f}_i(S_i) \right| = \sum_{1 \leq i \leq n} |\vec{f}_i(S_n)| = (n+1) \cdot n! = (n+1)!$$

For finite sets, we can say that $H \subseteq G$ is equal to G when $|H| = |G|$. Hence the union of all the products of permutations in S_n with a cycle in S_{n+1} is itself just S_{n+1} . $\square$

Theorem 1 (Permutations-Factorization). *Every permutation can be expressed as a product of cycles of length 2, which we call prime-factors of the permutation.*

Proof. Let's proceed by induction on the number of elements in the set A . The base case is when A has only two elements is trivially true, since the only permutations are the $(1, 2)$ cycle and the identity, which can be expressed as the product $(1, 2) \circ (2, 1)$.

Now, for every permutation in S_n assume that a factorization exists. Observe that that every permutation in S_n can be expressed as a product of length-cycles, and such product of cycles also describes a permutation in S_{n+1} .

Since by the result of lemma 8 we show that S_{n+1} can itself be described as the group of all the products of permutations in S_n with a length-2-cycle, every permutation in S_{n+1} is hereby the product of a sequence of length-2-cycles. $\square$

2.2 Normal subgroups

In this section, we will define the concept of normal subgroup and we will also talk about how normal subgroups are related with the concept of quotient group, and conjugacy classes. We will also define the concept of simple group and we will discuss the Lagrange's theorem as a mean to help us prove that a given group is simple.

Definition 17 (Normal subgroup). *For every H subgroup of G , we say that H is a normal subgroup of G , and we indicate it with $H \triangleleft G$ if it's closed under conjugation:*

$$\begin{aligned} H \triangleleft G &\iff \forall h \in H \quad \forall g \in G \quad ghg^{-1} \in H \\ H \triangleleft G &\iff \forall g \in G \quad gHg^{-1} = H \end{aligned}$$

Lemma 9. *If a group G has a normal subgroup H , then the left cosets of H in G are equal to the right cosets of H in G and viceversa.*

Proof.

$$\begin{aligned} gHg^{-1} &= H \\ gHg^{-1}g &= Hg \\ gH &= Hg \end{aligned}$$

□

Definition 18 (Conjugacy class). *The conjugacy class of an element $g \in G$ is the set of all elements that are conjugate to g in G , that is, the set of all hgh^{-1} :*

$$\text{Conj}_G(g) = \{hgh^{-1} \mid h \in G\}$$

Lemma 10. *Conjugacy classes are either completely disjoint or completely identical*

Proof.

$$\begin{aligned} \text{Conj}_G(a) \cap \text{Conj}_G(b) \neq \emptyset &\Rightarrow [\exists h, k \in G \quad hah^{-1} = kbk^{-1}] \\ hah^{-1} = kbk^{-1} &\Rightarrow k^{-1}hah^{-1} = bk^{-1} \Rightarrow (k^{-1}h)a(h^{-1}k) = b \\ hah^{-1} = kbk^{-1} &\Rightarrow ah^{-1} = h^{-1}kbk^{-1} \Rightarrow a = (h^{-1}k)b(k^{-1}h) \end{aligned}$$

□

Theorem 2 (Lagrange's theorem). *The order of H subgroup of G divides $|G|$.*

Proof. The left cosets in H , meaning the family of sets gH for some fixed $g \in G$, are all of the same size, since the value of g doesn't alter the number of elements in H . More specifically, the size of the left cosets is equal to the size of H .

$$a \sim b \iff a \in bH$$

We can easily see that the $\sim$ relation is an equivalence relation, thus, gH are a family of equivalence classes, hereby they partition G , hence the order of G is the sum of the orders of the left cosets (which all have the same size):

$$|G| = |g_1H| + |g_2H| + \dots = |H| + |H| + \dots = n \cdot |H|$$

□

Lemma 11. *The intersection of two normal subgroups is a normal subgroup.*

Proof. Let H_1 and H_2 be two normal subgroups of G . Then, by definition of normality:

$$\begin{aligned}\forall h \in (H_1 \cap H_2) \quad \forall g \in G \quad ghg^{-1} \in H_1 \wedge ghg^{-1} \in H_2 \\ \therefore ghg^{-1} \in (H_1 \cap H_2)\end{aligned}$$

□

Definition 19 (Simple group). *A group G is called simple if it has no non-trivial normal subgroups. In other words, the only normal subgroups of G are itself and $\{\epsilon\}$.*

Definition 20 (Quotient of groups). *The quotient $\frac{G}{H}$ is the set of all the left cosets of H in G . By this, we mean that for every fixed $g \in G$, gH must be an element of $\frac{G}{H}$.*

We are interested into studying the quotient of groups, in particular, we want to understand when the quotient of a groups is a group itself. To do so, we will first need to define an operation on elements of a quotient. Such operation is $\times$:

$$aH \times bH = (ab)H$$

This operation is well defined only if for every possible choice of $n_1, n_2 \in N$ exists and $n_3 \in N$ such that the following holds true:

$$\begin{aligned}(an_1)(bn_2) &= (ab)n_3 \\ (an_1)(bn_2) &\in (ab)N\end{aligned}$$

Theorem 3. *The quotient $(\frac{G}{H}, \times)$ is a group if and only if H is normal subgroup of G*

Proof.

$$\begin{aligned}(an_1)(bn_2) &\in (ab)N \\ (an_1)(bn_2) &= (ab)n_3 \\ an_1bn_2 &= abn_3 \\ n_1bn_2 &= bn_3 \\ b^{-1}n_1bn_2 &= n_3 \\ b^{-1}n_1b &= n_3n_2^{-1} \\ b^{-1}n_1b &\in N \\ b^{-1}Nb &= N\end{aligned}$$

□

2.3 Group homomorphisms

In this section, we will define the concept of group homomorphism, which we will later use to prove a result about the solvability of polynomial equations.

Definition 21 (Group homomorphism). *A group homomorphism is a function $\varphi : G \rightarrow H$ between two groups G and H such that:*

$$\forall g_1, g_2 \in G \quad \varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$$

Definition 22 (Kernel). *We call kernel of a group homomorphism $\varphi : G \rightarrow H$ the set of elements in G that are mapped to the identity element of H by φ .*

$$\ker \varphi = \{g \in G : \varphi(g) = 1_H\}$$

Definition 23 (Group epimorphism). *A group homomorphism $\varphi : G \rightarrow H$ that is also a surjective function is called a group epimorphism.*

Lemma 12. *The image of H normal subgroup of G via the epimorphism $\varphi : G \rightarrow K$ is a normal subgroup of the image of K via the same epimorphism ($H \triangleleft G \Rightarrow \vec{\varphi}(H) \triangleleft K$).*

Proof. Consider the group $\vec{\varphi}(H)$ and show that it is normal in $K = \vec{\varphi}(G)$:

$$\forall k \in K \quad \forall l \in \vec{\varphi}(H) \quad k l k^{-1} = \varphi(g) \varphi(h) \varphi(g^{-1}) = \varphi(ghg^{-1}) \in \vec{\varphi}(H)$$

This proof uses the fact that φ is an epimorphism, hence it is surjective, to conclude that for every k, l exists g, h such that $k l k^{-1} = \varphi(g) \varphi(h) \varphi(g^{-1})$. For non surjective homomorphisms, the proof is not valid, because there might be elements in K that are not images of elements in G , herby it could be impossible to find g, h .

□

Definition 24 (Group endomorphism). *A group homomorphism $\varphi : G \rightarrow H$ that is also a injective function is called a group endomorphism.*

Definition 25 (Group isomorphism). *A group homomorphism $\varphi : G \rightarrow H$ that is both injective and surjective (herby a bijection) is called a group isomorphism.*

Lemma 13. *If A is an abelian group and $\varphi : A \rightarrow B$ is a group isomorphism, then B is also an abelian group (notice that since a bijection is surjective $B = \vec{\varphi}(A)$).*

Proof. Since φ is a group isomorphism, it is a bijection, herby every element in $b_i \in B$ is the image of an element in $a_i \in A$ via φ , thus, the following holds true:

$$b_1 b_2 = \varphi(a_1) \varphi(a_2) = \varphi(a_1 a_2) = \varphi(a_2 a_1) = \varphi(a_2) \varphi(a_1) = b_2 b_1$$

□

Theorem 4 (First theorem of isomorphism). *For every group epimorphism $\varphi : A \rightarrow B$, the image of A via φ is isomorphic to the quotient group A divided by the kernel of φ :*

$$\frac{G}{\ker_\varphi} \sim \vec{\varphi}(G)$$

We can proof that these groups are in fact isomorphic by showing that the function ψ is an isomorphism between them, since it's clearly both injective and surjective.

$$\begin{aligned} \psi : \frac{G}{\ker_\varphi} &\rightarrow \vec{\varphi}(G) \\ \psi(g\ker_\varphi) &= \varphi(g) \end{aligned}$$

Theorem 5 (Second theorem of isomorphism). *Consider the groups $H \triangleleft B \triangleleft A$. Then, there is an isomorphism between the quotient groups (AH/BH) and $\frac{A}{B}$.*

$$\frac{AH}{BH} \sim \frac{A}{B}$$

Proof. To prove that exists an isomorphism between such groups, we will rely on the previous result of the first theorem of isomorphism (theorem 4). We therefore need to exhibit a homomorphism between the groups AH and $\frac{A}{B}$:

$$\begin{aligned} \varphi : (AH) &\rightarrow \frac{A}{B} \\ \varphi(ah) &= aB \end{aligned}$$

Constructed as such, we can see that φ is a homomorphism, since B is a normal subgroup of A , hence, left cosets and right cosets are equal (lemma 9).

$$\varphi(a_1h_1)\varphi(a_2h_2) = a_1Ba_2B = a_1B(a_2B) = a_1a_2BB = a_1a_2B = \varphi(a_1h_1a_2h_2)$$

We can also see that it is a surjective function, since for every aB in $\frac{A}{B}$, we can find ah in AH such that $\varphi(ah) = aB$. So let's go ahead and inspect it's kernel (the set of elements mapped to the identity). Please note that the kernel of this epimorphism is just BH :

$$\begin{aligned} \varphi(ah)\varphi(b_1h_1 \in BH \subseteq AH) &= aB = \varphi(ah) \\ \ker_\varphi &= BH \end{aligned}$$

We can herby apply the first theorem of isomorphism (theorem 4) to finish the proof:

$$\frac{AH}{BH} = \frac{AH}{\ker_\varphi} \sim \vec{\varphi}(AH) = \frac{A}{B}$$

□

Theorem 6 (Third theorem of isomorphism). *Consider the groups $H \triangleleft B \triangleleft A$. Then, there is an isomorphism between the quotient groups $(\frac{A}{H}/\frac{B}{H})$ and $\frac{A}{B}$.*

$$\frac{(\frac{A}{H})}{(\frac{B}{H})} \sim \frac{A}{B}$$

To prove this theorem, it's helpful to keep in mind the form of the generic element of the set (groups) we're working with (for an easier notation, we will call $H\frac{B}{H}$ by the name K):

$$\frac{A}{H} = \{aH : a \in A\}$$

$$\frac{B}{H} = \{bH : b \in B\}$$

$$\frac{A}{B} = \{aB : a \in A\}$$

$$\frac{(\frac{A}{H})}{(\frac{B}{H})} = \{aK : a \in A\}$$

Proof. To prove that exists an isomorphism between such groups, we will just exhibit it:

$$\begin{aligned} \varphi : (\frac{A}{H})/(\frac{B}{H}) &\rightarrow \frac{A}{B} \\ \varphi(aK) &= aB \end{aligned}$$

First things first: let's show that it is indeed an homomorphism (compatibility with $\cdot$):

$$\varphi(rK)\varphi(tK) = rBtB = r(Bt)B = r(tB)B = rt(BB) = rtB = (rt)B = \varphi(rtK)$$

Note that here in this very argument we used the all these subgroups are normal, hence every right coset is equal to a left coset (lemma 9) when writing $r(Bt)B = r(tB)B$.

We can clearly see that it is a bijection, since for every aB in $\frac{A}{B}$, we can find aK in $(\frac{A}{H})/(\frac{B}{H})$ such that $\varphi(aK) = aB$. Thus we only really need to show that it's an injection to complete the proof. Injectivity can be shown with the following argument:

$$\varphi(rK) = \varphi(tK) \Rightarrow rB = tB \Rightarrow r\epsilon = t\epsilon \Rightarrow r = t$$

These key observations show that the function φ is indeed a bijective homomorphism, hereby it is an isomorphism. To express the fact that an isomorphism exists between two groups the symbol $\sim$ is used (sometimes it can be used $\cong$ as well, but not here).

□

2.4 The alternating group (A_n)

The alternating group A_n is a very peculiar subgroup of the symmetric group S_n . It is defined as the group of "even" permutations of n elements. We will see how this group is special in many ways. It is the biggest normal subgroup of S_n and this will later turn out to be the key of the unsolvability of the general polynomial equations of degree $n \geq 5$, since we will discover that A_n being simple and non-abelian for $n \geq 5$ implies that S_n is not Galois-solvable for $n \geq 5$.

Definition 26 (Even permutation). *A permutation $\sigma \in S_n$ is called an even-permutation if it can be expressed as a product of an even number of length-2-cycles.*

Definition 27 (Alternating group). *The set of all even permutations of S_n is called the alternating group and is denoted by A_n . Since the product of two even permutations is an even permutation, the set A_n is a subgroup of S_n .*

Lemma 14. *The alternating group A_n is non-abelian for $n \geq 4$.*

Proof. We can exhibit an example of two even permutations that do not commute in A_4 , for example the permutations $(1, 2)(3, 4)$ and $(1, 3)(2, 4)$. Since these permutations will also be in A_n for every $n \geq 4$, we can conclude that A_n is non-abelian for $n \geq 4$. □

Lemma 15. *The alternating group A_n is a normal subgroup of S_n .*

Proof. For every $g \in S_n$ and $h \in A_n$, we can write the permutation g as a product of k length-2-cycles, while for every $h \in A_n$, we can write it as a product of $2m$ length-2-cycles:

$$\begin{aligned} g &= (a_1, b_1) \circ (a_2, b_2) \circ \cdots \circ (a_{k-1}, b_{k-1}) \circ (a_k, b_k) \\ h &= (x_1, y_1) \circ (x_2, y_2) \circ \cdots \circ (x_{2m-1}, y_{2m-1}) \circ (x_{2m}, y_{2m}) \end{aligned}$$

This means that the product ghg^{-1} can be written as the product of $k + 2m + k$ length-2-cycles, which is an even number, hence ghg^{-1} is an even permutation. Since ghg^{-1} is an even permutation, we can conclude that $ghg^{-1} \in A_n$. This means that A_n is closed under conjugation, hence it is a normal subgroup of S_n . □

Theorem 7. *If that A_n is simple, There is no other non-trivial normal subgroup of S_n*

Proof. Let H be a normal subgroup of S_n . Then, by lemma 11, we have that $H \cap A_n$ is a normal subgroup of A_n and since A_n is simple, we have that $H \cap A_n = \{\epsilon\}$ or $H \cap A_n = A_n$.

If $H \cap A_n = \{\epsilon\}$, then H is either the trivial subgroup or it is a group with only odd permutations. Since H must be a group, it must be closed under the product of permutations, so it must contain the product of two odd permutations, which is an even permutation. Hence H must contain an even permutation, which is a contradiction.

If $H \cap A_n = A_n$, then H contains A_n , so either they are identical or there is an element of S_n which is in H but not in A_n . Suppose that H is not identical to A_n . Then, the only element of H that is not in A_n must be an odd permutation α , but since H is a group, it must be closed under the product of permutations, so it must contain $\alpha(A_n)$ which is to say that it has to contain all products of even permutations with α .

So we have that H contains $\vec{f}(A_n)$ where f is a function that takes an even permutation and returns itself multiplied by α . The function is injective:

$$\begin{aligned} f(\sigma) &= f(\tau) \\ \sigma \circ \alpha &= \tau \circ \alpha \\ \sigma \circ \alpha \circ \alpha^{-1} &= \tau \circ \alpha \circ \alpha^{-1} \\ \sigma &= \tau \end{aligned}$$

Also, since $f(\sigma)$ is an odd permutation, we can say that $\vec{f}(A_n)$ disjoint from A_n . This means that H contains two disjoint sets of permutations, both of which are of cardinality $|A_n|$ which implies that $|H| = 2|A_n| = |S_n|$ Hence H is itself just S_n .

□

Lemma 16. *Every normal subgroup $H \triangleleft S_n$ that contains a length-3-cycle must contain all length-3-cycles given that $n \geq 3$.*

Proof. Assume $H \triangleleft S_n$ and that H contains a length-3-cycle (a, b, c) . Then consider that by definition of normality we have that the following statement holds true:

$$\forall \pi \in S_n \quad \pi \circ (a, b, c) \circ \pi^{-1} \in H$$

We want to show that H contains all length-3-cycles. So, for the arbitrary length-3-cycle (i, j, k) we will show that we can find a permutation π such that:

$$\begin{aligned} \pi \circ (a, b, c) \circ \pi^{-1} &= (i, j, k) \\ \pi(x) &= \begin{cases} a & \text{if } x = i \\ b & \text{if } x = j \\ c & \text{if } x = k \\ x & \text{otherwise} \end{cases} \end{aligned}$$

□

Lemma 17. *Every even permutation in A_n of length 2 can be expressed as a product of length-3-cycles for $n \geq 4$.*

Proof. We can express every even permutation of length 2 in the form $(a, b) \circ (b, c)$ as the product of two length-3-cycles $(a, b, c) \circ (1)$. So, without loss of generality, we can focus just on permutations of the form $(a, b) \circ (c, d)$.

We can easily show that every permutation in the form $(a, b) \circ (b, c)$ can be expressed as the product of multiple length-3-cycles as shown below in figures 2.4 and 2.4

□

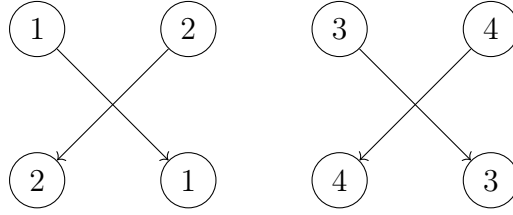


Figure 1: The permutation $[(1, 2) \circ (3, 4)]$

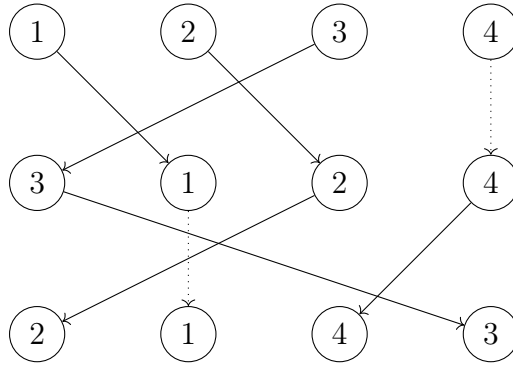


Figure 2: The permutation $[(1, 2, 3) \circ (1, 4, 3)]$

Lemma 18. *Every even permutation in A_n of arbitrary length can be expressed as a product of length-3-cycles for $n \geq 4$.*

Proof. By definition, every even permutation of length $2k$ is a product of $2k$ length-2-cycles, which are called prime-factors. We can consider them pairwise and define every even permutation as a product of k even permutations of length 2. Thanks to lemma 18 we can hereby express every even permutation of length $2k$ as a product of k length-3-cycles, thus proving the lemma.

□

3 Fields and automorphisms

In this section we will explain the concept of field extensions of a given field, we will define different types of field extensions (e.g. normal-extension, galois-extension, radical-extension) and we will see how they are related to a particular group called Galois group. We will hereby lay the foundation for the next chapter that will treat the connection between field extensions and solvability of polynomial equations.

3.1 Field extensions

A field extension is a way to enlarge a field by adding new elements to it. We will reason about the generic field F , but it's important to note that the the real goal here is to derive some general results that can be applied to the field of coefficients of a given polynomial extended by its roots.

Definition 28 (Field extension). *A field extension $F(r_1, r_2, \dots, r_n)$ is the smallest possible field that contains every element in F and the elements $r_1, r_2, \dots, r_n$.*

$$F(r_1) = \{a + br_1 : a, b \in F\}$$
$$F(r_1, r_2, \dots, r_n) = [F(r_1, r_2, \dots, r_{n-1})](r_n)$$

Definition 29 (Normal extension). *A field extension $F(r_1, r_2, \dots, r_n)$ is a normal extension of F if every polynomial $f(x) \in F[x]$ that has a root in $F(r_1, r_2, \dots, r_n)$ splits into linear factors in $F(r_1, r_2, \dots, r_n)$. In other words, a normal extension is a field extension that contains all the roots of every polynomial that has at least one root in it.*

$$F \subseteq E \text{ is normal} \iff \forall p \in F[x] \ \forall r \in E \ [p(r) = 0 \Rightarrow p(x) = (x - r) \cdot p'(x)]$$

Example 11. *It's difficult to prove, but hopefully not difficult to believe, that $\mathbb{C} = \mathbb{R}(i)$, where i is the imaginary unit, is a normal extension of $\mathbb{R}$. The reason is that every polynomial $p(x) \in \mathbb{R}[x]$ that has a root in $\mathbb{C}$ splits into linear factors in $\mathbb{C}$. For example, the polynomial $x^2 + 1$ splits into linear factors as $(x - i)(x + i)$.*

Definition 30 (Separable extension). *A field extension $F(r_1, r_2, \dots, r_n)$ is a separable extension of F if given an element of such extension $a \in F(r_1, r_2, \dots, r_n)$, the monic irreducible polynomial of lowest degree n for which a is a root has exactly n distinct roots.*

Example 12. *Notice that $\mathbb{Q}$ is an extension of itself, and it's separable, since for every possible value in $a \in \mathbb{Q}$ we have that the minimal monic irreducible polynomial is $x - a$, which has one root and it's of degree 1. The field $\mathbb{Q}(\sqrt{2})$ is also a separable, since the only minimal monic irreducible polynomial of degree two that has roots in it is $x^2 - 2$, which has two distinct roots, namely $\sqrt{2}$ and $-\sqrt{2}$.*

3.2 The Galois group

The Galois group is a group of automorphisms of a field extension that fixes the base field. The Galois group is a central concept in Galois theory, since it's precisely what the theory is built upon.

Definition 31 (Automorphism). *A field-automorphism for the field $(F, +, \cdot)$ is a bijective map $\phi : F \rightarrow F$ such that for all $a, b \in F$, the basic properties of a field are preserved, meaning that ϕ act as a linear mapping with respect to the ring operations:*

$$\begin{aligned}\phi(a + b) &= \phi(a) + \phi(b) \\ \phi(ab) &= \phi(a)\phi(b)\end{aligned}$$

Lemma 19 (Zero-lemma). *For any automorphism ϕ of a field $(F, +, \cdot)$, we have that:*

Proof.

$$\begin{aligned}\phi(0) &= \phi(0 \cdot a) = \phi(0)\phi(a) \\ \therefore \phi(0) &= 0\end{aligned}$$

□

Lemma 20 (Unit-lemma). *For any automorphism ϕ of a field $(F, +, \cdot)$, we have that:*

Proof.

$$\begin{aligned}\phi(a) &= \phi(1 \cdot a) = \phi(1)\phi(a) \\ \therefore \phi(1) &= 1\end{aligned}$$

□

Lemma 21. *All the automorphisms of a given field form a group under the operation of composition.*

Proof. Consider two automorphisms ϕ_1 and ϕ_2 of a field F . The composition $\phi_1 \circ \phi_2$ is also an automorphism of F , since it preserves the field operations:

$$\begin{aligned}(\phi_1 \circ \phi_2)(a + b) &= \phi_1(\phi_2(a + b)) = \phi_1(\phi_2(a) + \phi_2(b)) = \phi_1(\phi_2(a)) + \phi_1(\phi_2(b)) \\ (\phi_1 \circ \phi_2)(ab) &= \phi_1(\phi_2(ab)) = \phi_1(\phi_2(a) \cdot \phi_2(b)) = \phi_1(\phi_2(a)) \cdot \phi_1(\phi_2(b)) \\ \therefore \phi_1 \circ \phi_2 &\text{ is an automorphism of } F\end{aligned}$$

□

Definition 32 (Galois group). *The Galois group of a field extension E for the field F is the group of automorphisms of E that fix F . In other words, the Galois group is the group of all the automorphisms of E that leave every element of F unchanged.*

$$\begin{aligned}\text{Aut}(E) &= \{ \phi : E \rightarrow E \text{ automorphisms} \} \\ \text{Gal}_F(E) &= \{ \phi \in \text{Aut}(E) : \forall a \in F \ \phi(a) = a \}\end{aligned}$$

Lemma 22 (roots-permutation lemma). *If ϕ is an automorphism of a field extension E of F , meaning it's an element of the Galois group $\text{Gal}_F(E)$, then, for every polynomial $p \in F[x]$ the automorphism ϕ must map roots of p to other roots of p .*

Proof.

$$\begin{aligned}\phi(p(x)) &= \phi(a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0) \\ &= a_n \phi(x^n) + a_{n-1} \phi(x^{n-1}) + \cdots + a_0 \\ &= a_n \phi(x)^n + a_{n-1} \phi(x)^{n-1} + \cdots + a_0 \\ &= p(\phi(x)) \\ \therefore \phi(p(x)) &= p(\phi(x))\end{aligned}$$

□

The roots-permutation lemma is important, since it highlights a very interesting property of the Galois group, which is the fact that for every polynomial $p \in F[x]$ that has roots $r_1, r_2, \dots, r_n$ not contained in F , the Galois group $\text{Gal}_F(E)$ will permute the roots of p and will hereby be isomorphic to the group of permutations of n elements S_n .

Theorem 8 (Fundamental theorem of Galois theory). *Take a field extension E of a field F , then there exists a bijection between the intermediate fields $F \subseteq I \subseteq E$ and the subgroups of the Galois group $\text{Gal}_F(E)$, defined as follows:*

$$\begin{aligned}\mathcal{G} : \{I : F \subseteq I \subseteq E\} &\rightarrow \text{subgroups of } \text{Gal}_F(E) \\ \mathcal{G}(I) &= \text{Gal}_I(E)\end{aligned}$$

Proof. The function $\mathcal{G}$ is clearly injective, so we only need to show that it's surjective. To show that, we will consider the generic subgroup H of the Galois group $\text{Gal}_F(E)$, and the set I of all the elements of E that are fixed by every automorphism in H . We will show that I is in the domain of $\mathcal{G}$ such that $\mathcal{G}(I) = H$ by its very construction.

First, let's show that not only every element of I is fixed by every automorphism in H , but also that every product and sum of elements of I is still fixed by every automorphism in H , hereby showing that I is closed.

$$\begin{aligned}\forall a, b \in I \quad \forall \phi \in H \quad \phi(a \cdot b) &= \phi(a)\phi(b) = ab \\ \forall a, b \in I \quad \forall \phi \in H \quad \phi(a + b) &= \phi(a) + \phi(b) = a + b\end{aligned}$$

Since H is a subgroup of $\text{Gal}_F(E)$, we have that every ϕ in H fixes F , meaning that $F \subseteq I$. Thus I is a field extension of F that maps to H via $\mathcal{G}$ (Galois correspondence). □

A very interesting consequence of the fundamental theorem of Galois theory is that every subgroup of a Galois group $\text{Gal}_F(E)$ is still a Galois group of some smaller field extension.

Theorem 9. *As a consequence of the first theorem of isomorphism (theorem 4) we can claim that there is an isomorphism between $\text{Gal}_A(B)$ and $\text{Gal}_A(S)/\text{Gal}_B(S)$ given that $A \subseteq B \subseteq S$ and that B is actually a normal extension of the field A .*

Proof. Let's define an homomorphism ϕ from $\text{Gal}_B(S)$ to $\text{Gal}_B(A)$ as shown below:

$$\begin{aligned}\psi : \text{Gal}_A(S) &\rightarrow \text{Gal}_A(B) \\ \psi(\sigma : S \rightarrow S) &= \sigma : B \rightarrow B\end{aligned}$$

This construction of ψ makes it map an automorphism that fixes A in S to an automorphism that is still fixing A but restricted to B . We need to show that it's an epimorphism, meaning that it is an homomorphism and that it's surjective, to then be able to inspect it's kernel and apply the first theorem of isomorphism (theorem 4).

To show that ψ is an epimorphism, we start by showing that it is a homomorphism:

$$\psi(\sigma \circ \pi) = \sigma \circ \pi = \psi(\sigma) \circ \psi(\pi)$$

Since ψ is compatible with the composition operation, we can say that it is a well-defined homomorphism. Now we need to show that it is surjective. The normality of B as an extension of A is crucial to establish surjectivity here, because B is normal, B is the splitting field of some polynomial over A . By the **extension theorem for automorphisms**, every automorphism $\tau \in \text{Gal}_A(B)$ extends to some $\tilde{\tau} \in \text{Gal}_A(S)$.

$$\forall \tau \in \text{Gal}_A(B), \exists \tilde{\tau} \in \text{Gal}_A(S) \text{ such that } \tilde{\tau}|_B = \tau.$$

Thus, $\psi(\tilde{\tau}) = \tau$, proving ψ is surjective. If B wasn't normal in A , there would exist automorphisms of B over A that cannot extend to S , breaking surjectivity. For example, if B lacked a root required to define $\tilde{\tau}$, ψ would fail to map onto $\text{Gal}_A(B)$.

Established that we're working with an epimorphism, we can now inspect its kernel:

$$\begin{aligned}[\psi(\sigma \circ \gamma) = \sigma] &\iff \forall b \in B \quad \gamma(b) = b \\ \ker_\psi &= \text{Gal}_B(S)\end{aligned}$$

We can apply theorem 4 (first theorem of isomorphism) to the current setup obtaining:

$$\text{Gal}_A(B) = \vec{\psi}(\text{Gal}_B(S)) \sim \frac{\text{Gal}_A(S)}{\ker_\psi} = \frac{\text{Gal}_A(S)}{\text{Gal}_B(S)}$$

□

3.3 Galois extensions

A Galois extension is a field extension that is both normal and separable. There is a very close relationship between Galois extensions and the Galois group, hence why we waited until now to define it.

Definition 33 (Galois extension). *A field extension E of F is a Galois extension if it is both normal and separable.*

Lemma 23. *If I is a Galois extension of F , and if E is a Galois extension of I , then all automorphisms in $\text{Gal}_F(E)$ maps elements of I to elements of I .*

Proof. given that I is a Galois extension of F , I is normal thus every irreducible polynomial in $F[x]$ with at least one root in I splits completely into linear factors over I .

This guarantees that contains all roots of any element in ensuring automorphisms of E over I permute these roots, meaning they must map elements of I to elements of I .

Being separable, on the other hand, guarantees that every element is root of at least one polynomial (the minimal monic irreducible polynomial of such element). Thus every element in I must be permuted by the automorphisms of E over I . □

Lemma 24. *If E is a Galois extension of F , then given an extension I such that $F \subseteq I \subseteq E$, such extension can be a Galois extension of F only if its Galois group is a normal subgroup of the Galois group of E .*

Proof. Let I be a Galois extension of F . By definition, I is both normal and separable, thus every automorphism in $\text{Gal}_F(E)$ must map elements of I to elements of I by lemma 23 (notice that this doesn't mean that I is fixed by such isomorphisms).

$$\text{Gal}_I(E) = \{\phi \in \text{Aut}(E) : \forall a \in I \ \phi(a) = a\}$$

$$\text{Gal}_F(E) = \{\phi \in \text{Aut}(E) : \forall a \in F \ \phi(a) = a\}$$

For every automorphism $\phi \in \text{Gal}_I(E)$, consider its conjugate $\sigma \circ \phi \circ \sigma^{-1}$ in $\text{Gal}_F(E)$. Such automorphism, maps elements of I to elements of I via σ^{-1} , then maps the results to themselves since ϕ fixes I being an element of its Galois group, and finally maps them back in I using σ that, because of lemma 23, maps elements of I in I , hence $\text{Gal}_I(E) \triangleleft \text{Gal}_F(E)$ since its closed under conjugation. □

Lemma 25. *If $Gal_I(E)$ is a normal subgroup of $Gal_F(E)$, then any automorphism in $Gal_F(E)$ maps elements of I to elements of I (this doesn't imply that it fixes I).*

Proof. Let $Gal_I(E) \triangleleft Gal_F(E)$. By definition of normal subgroup, we have that the conjugate of every automorphism in $Gal_I(E)$ is also in $Gal_I(E)$ (herby it fixes I).

$$Gal_I(E) \triangleleft Gal_F(E) \iff \forall \phi \in Gal_I(E) \quad \forall \sigma \in Gal_F(E) \quad [\phi \circ \sigma \circ \phi^{-1} \in Gal_I(E)]$$

Notice that $\phi \circ \sigma \circ \phi^{-1}$ has to fix I for every choice of σ , and there are many possible σ that permute elements of $E - I$ in an unpredictable way. The only elements that are guaranteed to be fixed by σ are the ones in I . In particular, for our choice of ϕ , any possible σ that permutes elements of $E - I$ must not interfere with the fact that $\phi \circ \sigma \circ \phi^{-1}$ fixes I . The only way for this to be possible is if ϕ maps element of I to element of I herby avoiding unpredictable shuffling by σ .

To put it *more formally*, assume that $\phi^{-1}(i \in I) = e \in E - I$, then, by carefully choosing σ such that $\sigma(e) = e_2 \neq e$, and that's possible since σ doesn't have to fix I , we end up in a situation where $\phi \circ \sigma \circ \phi^{-1}(i) = \phi(e_2) \neq i$. This is a contradiction and it arises from the assumption that ϕ can map elements of I outside of I . Thus, that must be impossible. $\square$

Lemma 26. *If E is a normal extension of F , and if $Gal_I(E)$ is a normal subgroup of $Gal_F(E)$, then I is a normal extension of F .*

Proof. By the previous lemma (lemma 25) we know that every automorphism in $Gal_F(E)$ must map elements of I to elements of I .

Since we know that E is a normal extension of F , we know that every irreducible polynomial in $F[x]$ with at least one root in E splits completely into linear factors over E , meaning that E contains all roots of such polynomial.

All the roots of this polynomial are permuted by the automorphisms in $Gal_F(E)$, and since such automorphisms must map elements of I to elements of I , we know that I contains all roots of every polynomial with at least one root in I . $\square$

Theorem 10 (Galois extensions theorem). *If E is a Galois extension of F , and if I is a field extension of F such that $I \subseteq E$, then I is a Galois extension of the field F if and only if $Gal_I(E)$ is a normal subgroup of $Gal_F(E)$.*

Proof. The first implication is given by lemma 24, while the second is given by lemma 25 in conjunction with the key observation that separability is always preserved for every subfield of a separable field. Thus the proof is complete. $\square$

4 Solvability of polynomial equations

In this chapter we will finally prove the main result of this notes, which is the fact that there is no general method to solve polynomial equations of degree five or higher. We will do this by showing that the Galois group of a polynomial of degree five or higher is not "Galois-solvable", while also explaining how this is connected to the solvability of the general polynomial equation.

4.1 Galois-solvability of groups

A key concept in Galois theory is the notion of solvability of a group. In this chapter we will define what is a Galois-solvable group, and we will explore their properties.

Definition 34 (Galois-solvable group). *A Galois group G is said to be Galois-solvable if some of its normal subgroups can be arranged in a chain of the form:*

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

such that each factor group G_{i+1}/G_i is abelian for every choice of i . This of course implies that G_0 must be abelian, since $G_0/\{e\}$ is just G_0 itself. Since $\{e\} \triangleleft G$, if G were to be abelian the condition would be met, meaning that every abelian group is solvable.

Lemma 27. *For every $H \triangleleft G$ whenever G is solvable, then H is also solvable.*

Proof. Consider the following solvability chain for the group G that is solvable:

$$\{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft G_3 \triangleleft \cdots \triangleleft G_{n-2} \triangleleft G_{n-1} \triangleleft G$$

Consider $H_i = G_i \cap H$ for every i . We can see that H_i is normal in H since the intersection of normal subgroups is a normal subgroup. We only need to show that the factor groups H_{i+1}/H_i are abelian. So let's just apply the definition:

$$\{e\} = H_0 \triangleleft H_1 \triangleleft H_2 \triangleleft H_3 \triangleleft \cdots \triangleleft H_{n-2} \triangleleft H_{n-1} \triangleleft H$$

$$G_{i+1}/G_i = \{ xG_i : x \in G_{i+1} \}$$

$$H_{i+1}/H_i = (G_{i+1} \cap H)/(G_i \cap H)$$

$$H_{i+1}/H_i = \{ xH_i : x \in H_{i+1} \}$$

$$\forall x, y \in G_{i+1} [xyG_i = yxG_i] \Rightarrow \forall x, y \in H_{i+1} [xyH_i = yxH_i]$$

□

Theorem 11. *If G is solvable, the quotient G/H is also solvable for every choice of H .*

Proof. Consider the following solvability chain for the group G that is solvable:

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_i \triangleleft G_{i+1} \triangleleft G_{i+2} \triangleleft \cdots \triangleleft G$$

Just project the whole chain under the canonical projection $\pi(x) = xH$

$$\vec{\pi}(\{e\}) = \vec{\pi}(G_0) \triangleleft \vec{\pi}(G_1) \triangleleft \cdots \triangleleft \vec{\pi}(G)$$

Now, since $\pi(x) = xH$ is an epimorphism, we can use the lemma 12 to claim that the new chain is still a valid chain of subgroups. Let's study this projected chain further.

Please notice that $\vec{\pi}(\{e\})$ is just $eH = H$ with is solvable because it is a normal subgroup of the solvable group G (lemma 27), while $\vec{\pi}(G) = G/H$ is our quotient group.

$$\{e\} \triangleleft H_1 \triangleleft H_2 \triangleleft \cdots \triangleleft H \triangleleft \vec{\pi}(G_0) \triangleleft \vec{\pi}(G_1) \triangleleft \cdots \triangleleft \frac{G}{H}$$

Using theorems 5 and 6, we can inspect for isomorphisms between the groups of this construction, keeping in mind that isomorphisms preserves abelianity as in lemma 13.

$$\frac{\vec{\pi}(G_{i+1})}{\vec{\pi}(G_i)} = \frac{(G_{i+1}H/H)}{(G_iH/H)} \sim \frac{(G_{i+1}H)}{(G_iH)} \sim \frac{(G_{i+1})}{(G_i)}$$

We don't *really* need to worry about enforcng the condition of $H \triangleleft G_i \triangleleft G_{i+1}$, that is required for using the theorem 5, since if that wasn't the case, $\vec{\pi}(G_i) = H$, hence, we would already have arrived to the end of the solvability chain.

In case is not obvious why the previous statement is true, we can consider the image of $G_j \triangleleft H$ under the canonical projection $\pi(x) = xH$, hopefully this clarifies every doubt.

$$\vec{\pi}(G_j) = G_jH/H = HH/H = H/H = H$$

□

4.2 Solvability by radicals

Here we are, finally, proving once and for all that not all polynomial equations are solvable by radicals. To prove such result, our plan is the following: first we have to assume that given a polynomial with coefficients in an arbitrary field have n roots, then we extend the field with such roots, obtaining a *potentially* different splitting field through a normal extension, then we extend the normal extension up to the smallest radical extension possible. We will show that if extending the splitting field to a radical extension is possible, it means that the splitting field must have a solvable Galois group.

Definition 35 (Radical extension). *A radical extension $R = F(r_1, r_2, \dots, r_n)$ is an extension in which every r_i can be expressed as a power of some other element of a smaller radical extension of F . This sentence can be expressed in symbols as:*

$$(r_i)^k \in F(r_1, r_2, \dots, r_{i-1})$$

Definition 36 (Perfect field). *A perfect field, is a field with no two non-zero elements such that their product is zero. Such field is sometimes also sometimes called a zero-characteristic field or an integral field (rings are also sometimes called integral).*

Definition 37 (Root of unity). *In a field, a root of unity of order k is an element ω such that $\omega^k = 1$. Of course, 1 is always root of unity and it's called trivial.*

Lemma 28. *The Galois group $\text{Gal}_F(F(\omega))$ for some field F and some non-trivial root of unity ω is abelian, herby it's solvable.*

Proof. Consider the product (composition) of two automorphisms φ, ψ randomly chosen in $\text{Gal}_F(F(\omega))$. They commute:

$$\psi(\varphi(\omega^i)) = \psi(\omega^{ij}) = \omega^{ijk} = \omega^{ikj} = \varphi(\omega^{ik}) = \varphi(\psi(\omega^i))$$

□

Lemma 29. *The Galois group $\text{Gal}_{F(\omega)}(F(\sqrt[k]{\gamma}, \omega))$, for a perfect field F and a primitive root of unity ω , is cyclic and therefore abelian when $F(\sqrt[k]{\gamma}, \omega)$ is a normal extension.*

Proof. Consider the polynomial $x^k - \gamma$ with k distinct roots r_i . We know that every element in the form $\omega^i \sqrt[k]{\gamma}$ is a root of the polynomial $x^k - \gamma$. Since they are exactly k roots, we can conclude that this form describes all the roots of the polynomial.

Since automorphisms in $\text{Gal}_{F(\omega)}(F(\sqrt[k]{\gamma}, \omega))$ are defined as the automorphisms of $F(\sqrt[k]{\gamma})$ that fix everything in $F(\omega)$, we can conclude that the following " τ " is in such group:

$$\begin{aligned}\tau(\omega^r \sqrt[k]{\gamma}) &= \omega^r \tau(\sqrt[k]{\gamma}) \\ \tau(\sqrt[k]{\gamma}) &= \omega^j \sqrt[k]{\gamma}\end{aligned}$$

This means that, since ω is the primitive root of unity, and since the every non-trivial root of unity is a power of the primitive root, we can conclude that the generic automorphism φ in $\text{Gal}_{F(\omega)}(F(\sqrt[k]{\gamma}, \omega))$ is of the following form, herby generated from τ :

$$\varphi(\sqrt[k]{\gamma}) = (\tau \circ \tau \circ \dots \circ \tau)(\sqrt[k]{\gamma}) = \tau^t(\sqrt[k]{\gamma}) = \omega^{jt} \sqrt[k]{\gamma}$$

□

Theorem 12 (Fundamental theorem of polynomial equations). *Is possible to find a normal radical extension S of a perfect field F , such that S is splitting field for a given polynomial $p \in F[x]$ only if the Galois group of S is solvable.*

Proof. Let's proceed by induction on the number of distinct, linearly independent roots of the polynomial p that are not already in F . For $n = 1$, we have only one root γ , such that $S = F(\gamma)$ and ω be a primitive root of unity of order k such that γ^k is in F .

We know that both $Gal_F(F(\omega))$ and $Gal_{F(\omega)}(F(\gamma, \omega))$ are abelian, therefore they're solvable, due to lemmas 28 and 29 previously discussed. We can prove that $Gal_F F(\gamma, \omega)$ is solvable by exhibiting its solvability chain, which is composed of the following groups:

$$\{id_F\} \triangleleft Gal_{F(\omega)}(F(\gamma, \omega)) \triangleleft Gal_F(F(\gamma, \omega))$$

The group $Gal_{F(\omega)}(F(\gamma, \omega))$ is abelian, therefore its quotient with the trivial group also is abelian. The group $Gal_F(F(\gamma, \omega))$ on the other hand it's not guaranteed to be abelian, but the quotient $Gal_F(F(\gamma, \omega))/Gal_{F(\omega)}(F(\gamma, \omega))$ for sure is, since it is isomorphic to $Gal_F(F(\omega))$ due to theorem 9, which is known for being abelian.

$$\frac{Gal_F(F(\gamma, \omega))}{Gal_{F(\omega)}(F(\gamma, \omega))} \sim Gal_F(F(\omega))$$

We wanted to show that $Gal_F(S)$ must be solvable, that's not quite we wanted yet, but we're almost there. All we have to do is apply theorem 9 one more time:

$$Gal_F(S) = Gal_F(F(\gamma)) \sim \frac{Gal_F(F(\gamma, \omega))}{Gal_{F(\gamma)}(F(\gamma, \omega))}$$

Since $Gal_F(F(\gamma))$ is solvable, the whole quotient is, hereby $Gal_F(S)$ is solvable. Assuming the statement is true up to the n -th root, let's prove it for the case with polynomials of $n + 1$ roots. Let $\hat{S}$ be the splitting field of a polynomial with one root more than before, observe that $Gal_F(\hat{S})$ is still solvable because of, once again, theorem 9.

$$Gal_F(S) \sim \frac{Gal_F(\hat{S})}{Gal_{\hat{S}}(\hat{S})}$$

Since $Gal_F(S)$ and $Gal_{\hat{S}}(\hat{S})$ are solvable, it follows that $Gal_F(\hat{S})$ must also be solvable.

Notice that in this proof, we have used the fact that S is normal extension of F to be able to apply theorem 9 and the fact that S is radical to be able to apply lemma 29. Both of these assumptions were indeed necessary to complete the proof. □